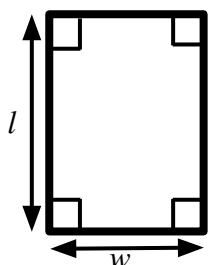




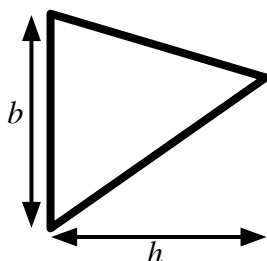
Problem solving with expressions and formulae

Task A: Forming expressions for area

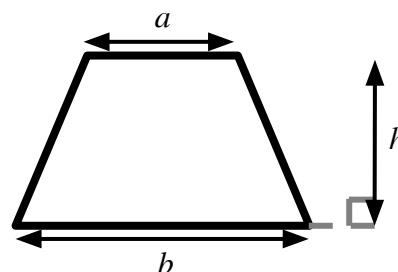
Here are some formula for the area of 2D shapes which may help you with the following task.



area = width $\times$ length



area = $\frac{1}{2} \times$ base $\times$ height



area = $\frac{1}{2} \times (a + b) \times$ height

Where a and b are the parallel sides.

1) Write a sentence explaining why the area of this shape can be written in each of the following ways:

a) $(x - 4)^2$

The shape is a square so the area can be found by squaring one side length.

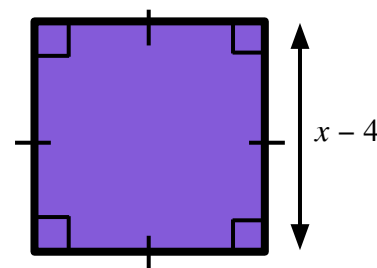
b) $(x - 4)(x - 4)$

Squaring an expression is the same as multiplying that expression by itself. $(x - 4)^2 \equiv (x - 4)(x - 4)$

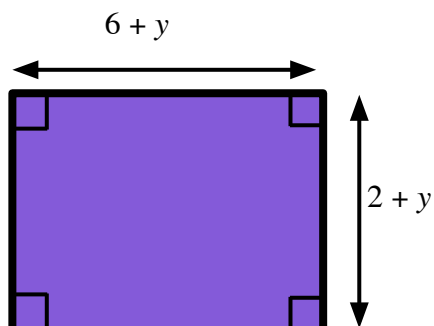
c) $x^2 - 8x + 16$

We can multiply the binomials using the distributive law:

$$(x - 4)(x - 4) \equiv x(x - 4) - 4(x - 4) \equiv x^2 - 4x - 4x + 16 \equiv x^2 - 8x + 16$$



2) Draw a rectangle with an area of $(6 + y)(2 + y)$ write a different expression for the area of your rectangle.



$(y + 6)(y + 2)$ or equivalent binomials.

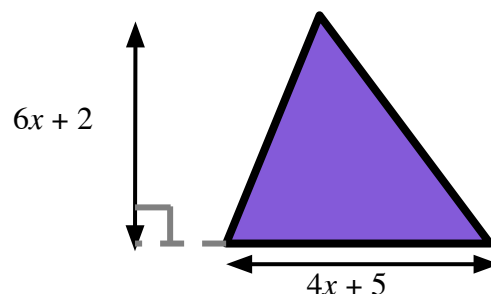
$$y^2 + 2y + 6y + 12$$

$$y^2 + 8y + 12$$

3a) Explain why the area of this shape be written as $\frac{1}{2} (4x + 5)(6x + 2)$

The area of a triangle can be calculated by multiplying the base by the height and multiplying by a half.

b) Write the expression for the area of this shape in two different ways.



E.g. $(4x + 5)(3x + 1)$

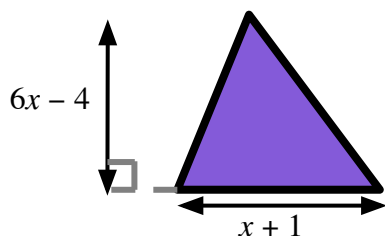
$(2x + 2.5)(6x + 2)$

$12x^2 + 4x + 15x + 5$

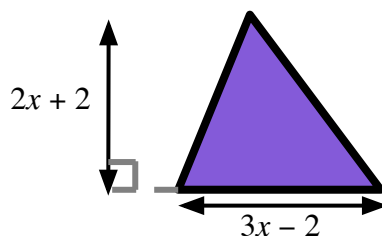
$12x^2 + 19x + 5$

$$\frac{(4x + 5)(6x + 2)}{2}$$

c) Draw and label a triangle with area $(x + 1)(3x - 2)$

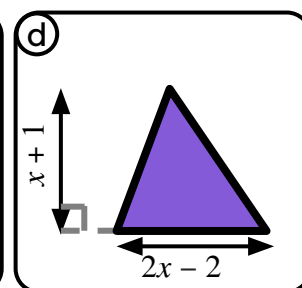
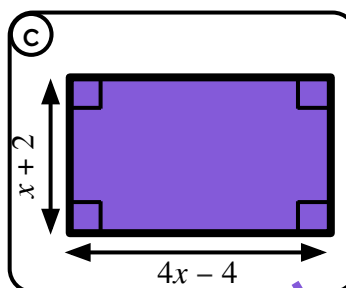
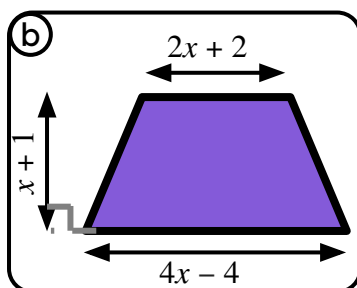
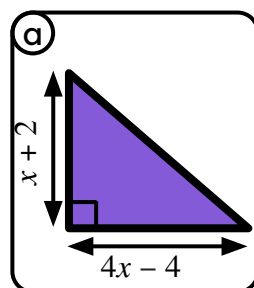


or



The binomials can be on either dimension.

4) Match these shapes to the correct expression for their area.



$\frac{1}{2}(4x - 4)(x + 2)$
 $(2x - 2)(x + 2)$
 $2x^2 + 4x - 2x - 4$

$\frac{1}{2}(6x - 2)(x + 1)$
 $(3x - 1)(x + 1)$
 $3x^2 + 3x - x - 1$

$(4x - 4)(x + 2)$
 $4x^2 + 8x - 4x - 8$

$\frac{1}{2}(2x - 2)(x + 1)$
 $(x - 1)(x + 1)$
 $x^2 + x - x - 1$

e) $x^2 - 1$

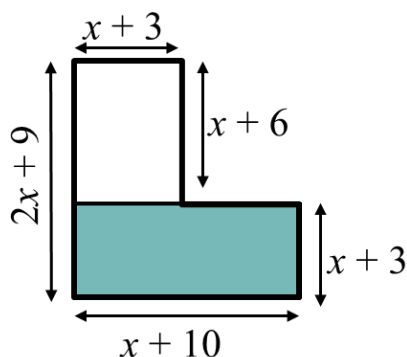
f) $2x^2 + 2x - 4$

g) $3x^2 + 2x - 1$

h) $4x^2 + 4x - 8$

Task B: Area of compound shapes

1) Calculate a simplified expression for the area of this shape.

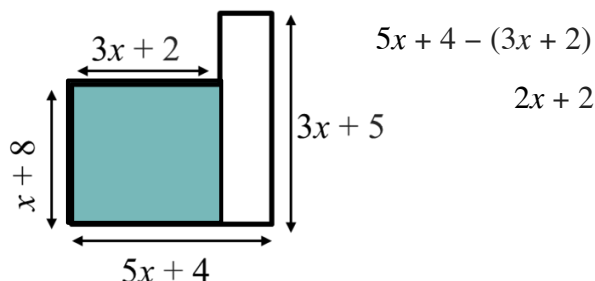


$$(x + 10)(x + 3) + (x + 6)(x + 3)$$

$$(x^2 + 13x + 30) + (x^2 + 9x + 18)$$

$$2x^2 + 22x + 48$$

2) Calculate a simplified expression for the area of this shape.



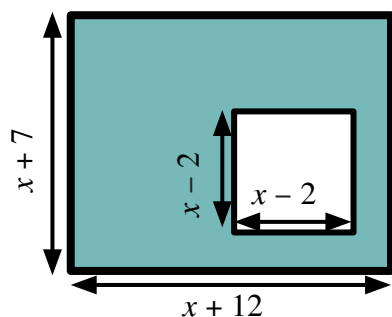
$$(3x + 2)(x + 8) + (2x + 2)(3x + 5)$$

$$(3x^2 + 24x + 2x + 16) + (6x^2 + 10x + 6x + 10)$$

$$(3x^2 + 26x + 16) + (6x^2 + 16x + 10)$$

$$9x^2 + 42x + 26$$

3a) Calculate a simplified expression for the area of the shaded shape.



$$(x + 12)(x + 7) - (x - 2)(x - 2)$$

$$(x^2 + 7x + 12x + 84) - (x^2 - 2x - 2x + 4)$$

$$(x^2 + 19x + 84) - (x^2 - 4x + 4)$$

$$23x + 80$$

b) The area of the shaded shape is 149 square units. Use this to find the value of x

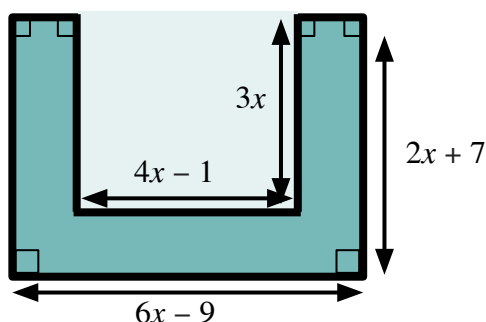
$$23x + 80 = 149$$

$$23x + 80 - 80 = 149 - 80$$

$$23x = 69$$

$$x = 3$$

4a) Calculate a simplified expression for the area of the shaded shape.



$$\begin{aligned}
 &(6x - 9)(2x + 7) - 3x(4x - 1) \\
 &(12x^2 + 42x - 18x - 63) - (12x^2 - 3x) \\
 &(12x^2 + 24x - 63) - (12x^2 - 3x) \\
 &27x - 63
 \end{aligned}$$

b) The area of the shaded shape is 99 square units. Use this to find the value of x

$$\begin{aligned}
 27x - 63 &= 99 \\
 (27x - 63) \div 9 &= 99 \div 9 \\
 3x - 7 &= 11 \\
 3x - 7 + 7 &= 11 + 7 \\
 3x &= 18 \\
 x &= 6
 \end{aligned}$$

5) Four identical rectangles sit round the outside of a shaded square.

a) Find a simplified expression for the area of one rectangle.

$$\begin{aligned}
 (2x + 3)(x + 2) &\equiv 2x^2 + 4x + 3x + 6 \\
 &\equiv 2x^2 + 7x + 6
 \end{aligned}$$

b) Find a simplified expression for the area of the whole shape.

$$2x + 3 + x + 2 \equiv 3x + 5$$

$$(3x + 5)^2 \equiv 9x^2 + 30x + 25$$

c) Find a simplified expression for the area of the shaded square.

$$4(2x^2 + 7x + 6) \equiv 8x^2 + 28x + 24$$

or

$$9x^2 + 30x + 25 - (8x^2 + 28x + 24) \equiv x^2 + 2x + 1$$

$$2x + 3 - (x + 2) \equiv x + 1$$

$$(x + 1)^2 \equiv x^2 + 2x + 1$$

